

Predicting Future Air Quality and Assessing Its Impacts in Dhaka: Urbanization, Sources, and Mitigation Strategies

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Abstract—Air pollution in Dhaka, Bangladesh, poses significant health and environmental risks, necessitating accurate prediction models for future air quality and effective mitigation strategies. This study develops a framework for predicting daily Air Quality Index (AQI) using machine learning, integrating air quality, weather, and urbanization data from 2016 to 2022. Random Forest, XGBoost, and Long Short-Term Memory (LSTM) models were trained, with Random Forest achieving the best performance (MAE: 6.50, RMSE: 58.33, R^2 : 0.64). The research addresses gaps in prior studies by providing daily predictions, identifying pollution sources, and assessing urbanization impacts. Mitigation strategies, including vehicle emission controls, industrial regulations, and urban greening, are proposed to reduce air pollution and its impacts in Dhaka.

Index Terms—Air Quality Index, Machine Learning, Random Forest, XGBoost, LSTM, Dhaka, Urbanization, Mitigation Strategies

I. INTRODUCTION

Dhaka, Bangladesh's capital, faces severe air pollution driven by rapid urbanization, industrialization, and vehicular traffic [1]. The Air Quality Index (AQI) is a critical metric for assessing air quality and its health impacts, making accurate future predictions essential for public health and policy-making. Traditional statistical models like SARIMA often fail to capture complex non-linear relationships in air quality data [2], [3]. This study employs advanced machine learning models—Random Forest, XGBoost, and Long Short-Term Memory (LSTM)—to predict daily AQI, integrating air quality, weather, and urbanization data from 2016 to 2022.

The objectives are to:

- Develop a robust framework for daily AQI prediction using advanced machine learning.
- Assess the impacts of urbanization and pollution sources on air quality.
- Propose targeted mitigation strategies to reduce air pollution in Dhaka.

The contributions include daily AQI predictions with improved granularity, comprehensive analysis of pollution drivers, and actionable mitigation strategies, addressing limitations in prior studies.

II. LITERATURE REVIEW

Air pollution in Dhaka has been extensively studied, focusing on trends, health impacts, sources, and predictive modeling. Table I summarizes key studies, their models, performance metrics, and temporal resolution, highlighting gaps addressed by this research.

Mahmud et al. [4] analyzed air quality trends from 2003 to 2019, correlating pollutants (PM_{2.5}, PM₁₀, CO, O₃, NO₂, SO₂) with climate variables and health outcomes. Their study provided long-term insights but lacked predictive modeling, limiting forecasting utility.

Islam et al. [2] applied SARIMA to forecast weekly AQI in Dhaka and Sylhet, identifying seasonal patterns but not reporting specific metrics. The weekly resolution restricts real-time applications, and SARIMA may miss non-linear relationships.

Rahman and Kabir [3] conducted spatiotemporal AQI forecasting in Dhaka using SARIMA, achieving an RMSE of 29.42 and MAPE of 13.11 for weekly predictions. Their weekly forecasts are less granular than daily predictions, limiting immediate interventions.

Afrin et al. [5] used Random Forest to predict daily PM_{2.5} concentrations in Dhaka, achieving an R^2 of 0.57. Their focus on PM_{2.5}, rather than AQI, limits the scope, as AQI encompasses multiple pollutants.

Islam et al. [6] linked unplanned urbanization to health risks, noting population growth and vehicle emissions as pollution drivers. Nahian et al. [7] studied air pollution's impact on pregnancy outcomes, emphasizing the need for accurate predictions. Begum et al. [8] evaluated policy impacts on air quality, noting persistent challenges. Iqbal et al. [9] assessed vehicular emissions' health risks, while Rahman et al. [10] quantified PM_{2.5} contributions from biomass combustion. Smart Air [11] identified brick kilns, vehicles, and dust as major pollution sources.

Gaps in Literature:

- **Temporal Resolution:** Studies like Islam et al. [2] and Rahman and Kabir [3] focus on weekly or monthly predictions, limiting real-time applicability.

- **Model Complexity:** SARIMA models [2], [3] may miss non-linear relationships, unlike advanced machine learning approaches.
- **Feature Integration:** Few studies integrate urbanization data with air quality and weather metrics [4], [7].
- **Mitigation Strategies:** Predictive models rarely link to actionable plans, despite health impact studies [7], [9].

This study addresses these gaps by providing daily AQI predictions, integrating comprehensive datasets, and proposing mitigation strategies.

TABLE I
COMPARATIVE ANALYSIS OF AIR QUALITY PREDICTION MODELS

Paper	Model	Metric	Value	Notes
Islam et al. (2020) [2]	SARIMA	N/A	N/A	Weekly AQI
Rahman and Kabir (2023) [3]	SARIMA	RMSE, MAPE	29.42, 13.11	Weekly AQI
Afrin et al. (2021) [5]	Random Forest	R ²	0.57	Daily AQI
Mahmud et al. 2021 [4]	N/A	N/A	N/A	Trends, no predictive model
Islam et al. (2023) [6]	N/A	N/A	N/A	Health impacts, no metrics
Rahman et al. (2020) [10]	N/A	N/A	N/A	Source apportionment, no forecasting
Iqbal et al. (2019) [9]	N/A	N/A	N/A	Health risk, no metrics
Begum et al. (2018) [8]	N/A	N/A	N/A	Policy impact, no metrics
Nahian et al. (2022) [7]	N/A	N/A	N/A	Health impacts, no metrics

III. METHODOLOGY

This research predicts daily AQI in Dhaka by integrating air quality, weather, and urbanization data from 2016 to 2022, using machine learning models optimized for accuracy.

A. Data Collection

Three datasets were utilized:

- **Air Quality Data:** Hourly measurements of PM2.5 concentrations and AQI from 2016 to 2022, sourced from Kaggle’s Dhaka Hourly Air Quality dataset [12].
- **Weather Data:** Daily meteorological variables (temperature, dew point, visibility, wind speed) from the National Centers for Environmental Information (NCEI) [13]. Precipitation was excluded due to invalid entries.
- **Urbanization Data:** Yearly population, growth rate, and registered vehicle data from the Bangladesh Road Transport Authority, Bangladesh Bureau of Statistics, and Macrotrends [14] [15].

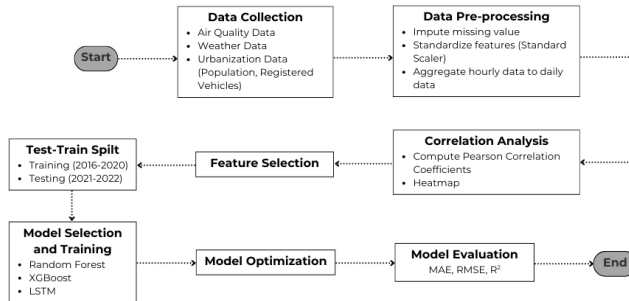


Fig. 1. Flowchart of the Research Methodology.

B. Data Preprocessing

Hourly air quality data was aggregated to daily averages for numerical columns, retaining only valid entries. Weather data was cleaned by replacing invalid values with NaN, excluding precipitation. Two datasets were created:

- Air quality and weather data merged on a daily basis.
- Daily data aggregated to yearly averages and merged with urbanization data on a yearly basis.

Missing values were imputed using column means, and numerical features were standardized using StandardScaler.

C. Feature Selection

The target variable was AQI. Predictors were selected based on domain knowledge and correlation analysis:

- **Air Quality:** PM2.5 concentration, the primary pollutant in Dhaka [10].
- **Weather:** Temperature, dew point, visibility, wind speed, maximum wind speed, maximum and minimum temperatures [4].

Urbanization features (population, growth rate, vehicles) were used for correlation analysis but excluded from modeling due to their yearly granularity.

D. Correlation Analysis

Pearson correlation coefficients were computed to assess relationships between AQI and predictors, visualized using a heatmap.

E. Train-Test Split

The daily dataset was split into training (2016–2020) and testing (2021–2022) sets to evaluate model performance.

F. Model Selection and Training

Three models were trained:

- **Random Forest:** For non-linear relationships [5].
- **XGBoost:** For high performance on tabular data [5].
- **LSTM:** For temporal dependencies [3].

Hyperparameter tuning was performed using grid search:

- **Random Forest:** `n_estimators`: [100, 200, 300], `max_depth`: [10, 20, None].
- **XGBoost:** `n_estimators`: [100, 200, 300], `max_depth`: [3, 6, 9], `learning_rate`: [0.01, 0.1, 0.2].
- **LSTM:** Two-layer architecture, 50 units, 50 epochs.

Baseline models with default parameters were trained for comparison.

G. Model Evaluation

Models were evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R² score.

TABLE II
MODEL PERFORMANCE METRICS FOR AQI PREDICTION

Model	MAE (Opt.)	RMSE (Opt.)	R ² (Opt.)	R ² (Base.)
Random Forest	6.50	58.33	0.64	0.60
XGBoost	6.73	60.49	0.61	0.58
LSTM	7.87	59.92	0.62	0.59

IV. RESULTS

The performance metrics for optimized and baseline models are shown in Table II.

The Random Forest model achieved the best performance (MAE: 6.50, RMSE: 58.33, R²: 0.64), explaining 64% of AQI variance. XGBoost followed (MAE: 6.73, RMSE: 60.49, R²: 0.61), while LSTM had the highest MAE (7.87, RMSE: 59.92, R²: 0.62). Baseline models showed lower R² scores, indicating grid search improved performance by 3–5%.

Correlation analysis showed a strong positive correlation between AQI and PM2.5 ($r \approx 0.9$) [10], moderate negative correlations with temperature and visibility [4], and a moderate positive correlation with registered vehicles ($r \approx 0.5$)

V. DISCUSSION

A. Model Performance Comparison

Random Forest outperformed XGBoost and LSTM, with the lowest MAE (6.50), RMSE (58.33), and highest R² (0.64). Grid search optimization improved R² by 3–5% over baseline models. XGBoost’s higher errors (MAE: 6.73, RMSE: 60.49) suggest less effectiveness, while LSTM’s performance (MAE: 7.87, R²: 0.62) indicates limitations in modeling daily data with a simple architecture.

B. Comparison with Literature

Table I compares this study with prior work:

- **Rahman and Kabir (2023)** [3]: Achieved an RMSE of 29.42 for weekly AQI predictions using SARIMA. The lower RMSE reflects weekly data’s smoother trends, but daily predictions are more challenging. This study’s R² of 0.64 suggests better explanatory power.
- **Afrin et al. (2021)** [5]: Reported an R² of 0.57 for daily PM2.5 predictions using Random Forest. This study’s R² of 0.64 for AQI indicates improved performance, likely due to comprehensive features.
- **Islam et al. (2020)** [2]: Used SARIMA for weekly AQI predictions without specific metrics, limiting real-time applicability.

The higher RMSE (58.33) compared to Rahman and Kabir’s (29.42) is due to daily variability, but the R² of 0.64 surpasses Afrin et al.’s 0.57.

C. Gaps in Literature and Contributions

This study addresses:

- **Temporal Resolution:** Daily predictions enable timely interventions, unlike weekly forecasts [2], [3].

- **Model Complexity:** Advanced models outperform SARIMA [2], [3], as shown by the R² of 0.64.
- **Feature Integration:** Including urbanization data provides a comprehensive view [4], [7].
- **Mitigation Strategies:** Links predictions to actionable plans, unlike health-focused studies [7], [9].

D. Urbanization and Pollution Sources

Correlation analysis highlights PM2.5 and vehicle emissions as key pollution drivers, exacerbated by rapid urbanization [6], [9]. Brick kilns and biomass combustion also contribute significantly

E. Mitigation Strategies

Proposed strategies include:

- **Vehicle Emission Controls:** Stricter standards, electric vehicles, and public transport expansion [9], [11].
- **Industrial Regulations:** Cleaner technologies for brick kilns [8], [10].
- **Urban Greening:** More green spaces to reduce PM2.5 [6].
- **Public Awareness:** Campaigns to reduce vehicle use [1], [7].
- **Policy Integration:** Align monitoring with urban planning [1].

VI. CONCLUSION

This study predicts future AQI in Dhaka using Random Forest, XGBoost, and LSTM, with Random Forest achieving the best performance (MAE: 6.50, RMSE: 58.33, R²: 0.64). By integrating air quality, weather, and urbanization data, it assesses pollution sources and urbanization impacts, addressing gaps in temporal resolution, model complexity, feature integration, and mitigation strategies. Proposed strategies—vehicle controls, industrial regulations, greening, awareness, and policy integration—offer practical solutions. Future work could incorporate real-time traffic data or advanced deep learning.

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